

AKSHAY BAWALIWALE

Senior Generative AI / LLM Engineer | Agentic AI, RAG, Voice AI, LLMOps

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SUMMARY

Generative AI & Machine Learning Engineer with **2+ years** shipping **20+ production LLM applications: multi-agent systems** (LangGraph, CrewAI), **RAG pipelines**, and **Voice AI**. Built systems handling **5,000 requests/day** with **92% top-k retrieval accuracy** and **50% cost reduction**. Equally strong in **applied machine learning** — building and benchmarking end-to-end systems from **recommendation engines** to **deep-learning computer-vision pipelines**.

TECHNICAL SKILLS

LLM & Agentic Systems:	LangChain, LangGraph, CrewAI, Tool Calling, Structured Outputs, ReAct Agents, MCP
RAG & Retrieval:	LlamaIndex, Pinecone, FAISS, ChromaDB, Embeddings, Retrieval Evaluation, RAGAS
LLMOps & Backend:	FastAPI, MLflow, LangSmith, CI/CD, Git, Model Evaluation and Benchmarking
Voice & Cloud:	LiveKit, Twilio SIP, Salesforce, AWS S3
Machine Learning:	Python, SQL, Scikit-learn, XGBoost, PyTorch, Hugging Face
Also Familiar With:	RLHF, PEFT, Prompt Optimization, Guardrails AI, A/B Testing, Google Vertex AI, Azure, R, SciPy, AWS CodeCommit

EXPERIENCE

Senior Generative AI & LLM Specialist

Jan 2026 – Present · Remote

BuzzBoard

- **Designed and deployed IRIS**, a three-agent **Voice AI workflow** integrating **LiveKit, Twilio SIP, and Salesforce**, reducing request-queue latency from **two days to three minutes (99.9%)**, and fulfillment time by **87.5%**, with real-time guardrails aligned with **HIPAA and PCI DSS** requirements.
- Led end-to-end **LLM evaluation and LLMOps** across response quality, latency, cost, and tool-use reliability using a **20-parameter evaluation framework** and **RAGAS**.
- Engineered production-grade **RAG content pipelines** that achieved **92% top-k retrieval accuracy on a curated evaluation set** and reduced manual review effort by **50%** through automated evaluation and quality-control workflows.
- Built and maintained **CI/CD pipelines on AWS** for scalable AI application deployments using **FastAPI, Git-based version control, and MLflow**, enabling continuous performance, experiment, and cost monitoring.
- Mentored **three engineers** across agentic system design, production deployment, evaluation frameworks, and engineering best practices.

Generative AI & LLM Specialist

Apr 2024 – Dec 2025 · Remote

BuzzBoard

- Architected and delivered **20+ production-grade Agentic AI systems** for BuzzBoard's US SMB client base, including **multi-agent BrandKit generation, Voice AI customer service, conversational agents, and social-content pipelines**, owning the complete lifecycle from architecture and development to deployment and production handover.
- Built a **multi-agent B2B sales-intelligence platform** using **ReAct-based Search, Knowledge, and Web agents**, reducing prospect research and lead-conversion cycles from **months to days** and cutting manual research effort by **75%**.

Full-Stack Data Science Trainee

Apr 2023 – Mar 2024 · Remote

AlmaBetter (Training Program)

- Completed an **Advanced Full-Stack Data Science and Machine Learning** program, with **800+ hours** of hands-on coding across **Python, SQL, EDA**, data preprocessing, **statistical analysis**, **feature engineering** and **predictive modeling**.
- Delivered **50+ end-to-end data science and machine learning projects** and assignments, covering problem formulation, data cleaning, visualization, **model development**, **hyperparameter tuning**, performance evaluation and business insight generation.

Research Staff

Jul 2022 – Oct 2022 · Remote

Centre of Excellence in AI, IIT Kharagpur

- Developed **image-processing and preprocessing algorithms** and contributed to a **U-Net-based semantic segmentation pipeline** for satellite imagery annotation that achieved **86% model accuracy**.

FEATURED PROJECTS

Netflix AI Recommender System [GitHub](#)

Python, Scikit-learn, K-means, NMF, LDA, Flask

- Built a Netflix recommendation system using **unsupervised clustering** (K-means, NMF, LDA), benchmarking **11 models** across a catalog of **8,807 titles** to improve content diversity by **43%** over single-metric baselines.
- Identified and corrected a high-accuracy model that produced a repetitive **genre echo chamber**, prioritizing balanced, relevant recommendations over a single performance metric. Added **cold-start handling** for new users and **explainable outputs** for transparency.

AI-Powered Image Super-Resolution API [GitHub](#)

PyTorch, Real-ESRGAN, Flask

- Engineered and deployed a production-ready image super-resolution API using **Real-ESRGAN** and **RRDBNet**, delivering **4× image upscaling** while preserving fine visual details. Optimized inference for resource-constrained environments using **tile-based processing**, **FP16 acceleration**, **thread-safe model loading**, and **automatic GPU/CPU fallback**.

EDUCATION

M.Tech, Earth System Science and Technology — IIT Kharagpur (2019–2021)

B.E., Mechanical Engineering — SSGI, Bhilai (2012–2016)

CERTIFICATIONS & WRITING

Reinforcement Fine-Tuning LLMs with GRPO — DeepLearning.AI

Machine Learning Specialization — Stanford University & DeepLearning.AI

Technical articles on **generative AI**, **agentic engineering**, and **LLM reinforcement learning** — Medium: [@akshay.bawali09](#)